

COMP232 Individual coursework

Assignment 1

Dr Anish Mukherjee

Overall marking scheme

The coursework for COMP232 consists of two assignments, with the contribution of each as follows:

- Assignment 1 - 15%
- Assignment 2 - 15%

This document describes Assignment 1 which will be marked according to the following general criteria:

- Correctness of the arguments.
- Presence/absence of the evidence on the experiments.
- Original contribution either in implementation, or analysis.

Failure in any assignment may be compensated by higher marks in other module components.

1 Estimation of the Time Required for a Brute-Force Search Attack on Password-Based Encryption

You are expected to have completed the tasks in Lab 1, Lab 2, and Lab 3.

1. For the password-based DES encryption implementation in JCA or Python (similar to Lab 2 and Lab 3), use a fixed salt and iteration count. For each password listed in Lab 1, Section 3, record the average time required for encryption and decryption (as in Lab 3).
2. For each of the passwords above, estimate the time required for a successful brute-force search attack, assuming that the attacker knows:
 - the predefined plaintext,
 - the resulting ciphertext,
 - the salt,
 - the iteration count,
 - but **not the password**.
3. Analyse how the time required for the attack depends on the iteration count.
4. Consider a variant in which the attacker has the knowledge listed above but does not know the iteration count. Estimate the time required to recover the passwords.

5. Compare your estimated times with those returned for the same passwords by the online services mentioned in Lab 1, and propose plausible explanations for any observed differences.

Write a report addressing the tasks above, providing *evidence* to support your arguments (e.g. snippets of code used to estimate the time required for an encryption). Aim to be *clear*, *precise*, and *succinct*.

As a general guideline, a clear and complete report for this assignment would typically be around 1,000–1,500 words, and reports within this range are entirely sufficient. This guidance is indicative only, and there is no minimum word count. Reports that are significantly longer are unlikely to be necessary and will not receive additional credit; clarity, relevance, and conciseness are valued over length. Tables, figures, code snippets, and references are *not included* in this guideline.

Note

- The work should be formatted as a lab report. If you are unsure how to structure or write a report, see: <https://liverpool.instructure.com/courses/24284/pages/academic-writing> or the relevant Canvas page.
- Note that the assignment tasks above *do not assume* that you will implement a full brute-force search program. You may wish to revisit the lab sheets and the simple Java or Python programs implementing password-based encryption used in Labs 2 and 3.
- JCA Reference Guide can be found at: <http://java.sun.com/javase/6/docs/technotes/guides/security/crypto/CryptoSpec.html>
- Please refer to pages 3 and 4 for the detailed marking descriptor.

Submission

- The report must be submitted via **Canvas** in PDF, DOC, or DOCX format.
- **Deadline: 17:00 on 13 March 2026.**

Please note that the standard University policies set out in the [Code of Practice on Assessment \(CoPA\)](#) apply to this assignment. In particular, please refer to Section 9 for policies on plagiarism, collusion, and fabricated data, and to Section 6 for policies on late submission and associated penalties.

Grade%	Descriptor
0-9%	No evidence of the use of a suitable methodology. Virtually devoid of any evidence of knowledge or understanding of the subject. No or almost no arguments/answers.
10-19%	Little evidence of the use of a suitable methodology. Shows only the most limited and fragmentary of the subject with little or no understanding of essential principles and concepts. Work is likely to be unstructured and ill-presented. Arguments/answers are only loosely related to issues/questions or only cover a seriously inadequate part of the issues/questions.
20-29%	Fails to demonstrate an ability to solve simple, well-defined, problems of a familiar type under guidance. Serious lack of the use of a suitable methodology or flaws in its use which negatively impact on the result of the work. Very limited range of knowledge with many important gaps and omissions. Shows incomplete understanding with numerous errors of interpretation. Arguments/answers have little structure, contains serious errors, and there is no support for arguments.
30-34%	Fails to demonstrate an ability to solve simple, well-defined problems of a familiar type. Lack of the use of a suitable methodology or flaws in its use which negatively impact on the result of the work. Knowledge and understanding of the subject are fragmentary, with an insufficient number of aspects showing a very basic level of understanding and too many aspects displaying fundamental errors omissions. Arguments/answers are lacking in structure.
35-39%	Fails to demonstrate an ability to solve limited, well-defined, problems of a familiar type. Aspects of a suitable methodology evident, but flaws in its use or omissions which negatively impact on the result of the work. Knowledge and understanding of the subject are fragmentary, some aspects showing a very basic level of understanding but other aspects displaying fundamental errors. Arguments/answers are lacking in structure. There are errors and omissions and the evidence provided to support arguments is very limited.
40-49%	Demonstrates an ability to solve limited, well-defined problems of a familiar type. Most aspects of a suitable methodology evident, but minor flaws in its use or omissions with some negative impact on the result of the work. General knowledge and understanding of the subject but very limited in depth or breadth. Arguments/answers are likely to be somewhat lacking in structure. There are likely to be errors and omissions and the evidence provided to support arguments will be very limited.

Grade%	Descriptor
50-59%	Displays ability to analyse, interpret and organise information to produce coherent accounts or solve well-defined problems of some scope. Most aspects of a suitable methodology evident and used correctly, some omissions occur but without negative impact on the result of the work. Satisfactory knowledge and understanding of the essentials of the subject, with an ability to integrate information into a clear, well-structured account, but lacking in breadth or depth, or with some significant aspects omitted. Arguments/answers must be clear, although they may not be well-developed or reflect a wider appreciation of the subject. Some errors and omissions are likely to be present.
60-69%	Demonstrates ability to analyse, interpret and organise information to produce coherent accounts or solve relatively complex problems. Use of a suitable methodology evident and used correctly, with minor omissions. Good knowledge and understanding of the subject, with no major gaps or omissions, but minor gaps or omissions may occur. Arguments/answers will be clear, competently structured, and largely accurate.
70-79%	Demonstrates ability to analyse, interpret and organise information to produce coherent accounts or solve complex problems. All aspects of a suitable methodology evident and used correctly. Comprehensive knowledge and understanding of the subject together with the ability to put the work into context and to critically evaluate selected aspects of the work. Arguments/answers will be clear, competently structured, and accurate.
80-89%	Displays a level of originality and creativity and/or the ability to suggest realistic solutions to novel problems. The methodology employed is well-developed and correct. Evidence of wide reading. For level 6 this should include relevant research papers and books. Perceptive, focused treatment of all issues/questions presented in a critical and scholarly way.
90-100%	Displays an exceptional degree of originality and creativity and/or exceptional analytical and problem-solving skills. The solution must have novel aspects. The methodology employed is well-developed and correct. Shows critical understanding of current knowledge. For level 6 this should include relevant recent research papers. Perceptive, focused treatment of all issues/questions presented in a critical and scholarly way.